

Syllabus for

Academic year 2021/2022 >

TIF075 - Environmental physics

Miljöfysik

Syllabus adopted 2021-02-26 by Head of Programme (or corresponding)

Owner: [TKTFY](#)

4,5 Credits

Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: First-cycle

Main field of study: Engineering Physics

Department: 16 - PHYSICS

Teaching language: Swedish

Application code: 57118

Open for exchange students: No

Maximum participants: 130

Only students with the course round in the programme plan

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course ¹	No	Sp	Examination dates ¹
0105 Project 4,5 c Grading: TH					4,5 c			

In programs

[TKTFY ENGINEERING PHYSICS, Year 2 \(compulsory\)](#)

Examiner:

[Dinko Chakarov](#)



[Go to course homepage](#)

Theme:

Environment 4,5 hec

Eligibility

General entry requirements for bachelor's level (first cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

The same as for the programme that owns the course.

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Course specific prerequisites

Previous courses in mathematics and physics within the programme

Aim

To study environmental issues and sustainable development in the context of physics.

Learning outcomes (after completion of the course the student should be able to)

- make simpler resource and sustainability analyses - describe and explain the global energy balance- explain the principles of solar, nuclear, water and wind power - explain how physical principles are connected to resource management and recycling issues - both orally and in writing convince about sustainability using proper analysed and balanced arguments - search for information and evaluate that information, available through Chalmers' library system.

Content

Presentation of different environment analysis systems. Project work on the sustainability of large scale technological systems, preferably energy-matter systems. Lectures on environmental problems and phenomena from a physics standpoint. Argumentation incl. technique in oral and writing. Mechanics and radiation applied to Earth's air and water problems. Energy and radiation balance concerning the climate. Diffusion and transport of pollutants in the atmosphere and in the oceans. About disturbances in environmental systems and human

activities. The course contains a compulsory information retrieval part, given by personnel at Chalmers library.

Organisation

A project work is running all through the course and in parallel there are lectures and exercises. Compulsory information retrieval part.

Literature

The main course literature will be announced on the course homepage prior to course start.

Examination including compulsory elements

Project work and home assignments. Passing the information retrieval part is required.

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers on educational support due to disability.